

Cycle 18 Autonomous Report

Date: 2026-05-16

Repository: qc_exciton_LCC

Task selected

Extend the autonomous-cycle PDF report generator so each cycle can render task-specific LaTeX-style equations instead of only the fixed default synthesis formulas.

Major changes

Added repeatable `--equation` arguments to `scripts/autonomous_cycle_report.py` while preserving the existing default equations.

Updated the smoke test to generate a PDF with custom mathtext equations.

Files changed

`scripts/autonomous_cycle_report.py`

`tests/test_autonomous_cycle_report.py`

`docs/cycle_reports/cycle18_report.pdf`

Rendered Equations

This page is a rendering check for the inbox requirement that equations appear as LaTeX-style math.

$$T_{cycle}(x) = T_{default}(x) \cup T_{task}(x)$$

$$\alpha_T < 1 \Rightarrow \text{sublinear QROAMClean amortization}$$

$$I_1(M, N) = T(M, N, 1, 0)$$

Goals already achieved

Reusable PDF reports now support task-specific rendered equations.

Prior cycles already achieved synthesis analytic-vs-Bloq Toffoli and qubit-count guards over the tabulated grid.

Verification

Tests and checks

PYTHONPATH=src python tests/test_autonomous_cycle_report.py -> 1/1 passed.

python -m py_compile scripts/autonomous_cycle_report.py
tests/test_autonomous_cycle_report.py -> passed.

Achieved goal

Improved report-generation infrastructure for final Toffoli complexity, qubit count, and scaling work while satisfying the inbox PDF requirement.

Next recommended task

Resume deriving $I_1(n_blocks, n_rows)$ in closed form from QROAMClean optimizer behavior, using scripts/regenerate_synthesis_tables.py as the regression oracle.